HOMEWORK 04

Name: **Student ID:**

1. (a) What angle in radians is subtended by an arc 1.50 m long on the circumference of a circle of radius 2.50 m? What is this angle in degrees? (b) An arc 14.0 cm long on the circumference of a circle subtends an angle of 128°. What is the radius of the circle? (c) The angle between two radii of a circle with radius 1.50 m is 0.700 rad. What length of arc is intercepted on the circumference of the circle by the two radii?

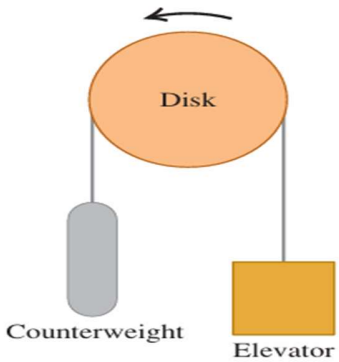
2. An airplane propeller is rotating at 1900 rpm (rev/min) (a) Compute the propeller's angular velocity rad/s. (b) How many seconds does it take for the propeller to turn through 35°?

3. A bicycle wheel has an initial angular velocity of 1.50 $^{rad}/_{s}$. (a) If its angular acceleration is constant and equal to 0.300 $^{rad}/_{s^2}$, what is its angular velocity at t = 2.50 s? (b) Through what angle has the wheel turned between t = 0 and t = 2.50 s?

4. An electric fan is turned off, and its angular velocity decreases uniformly from 500 $^{rev}/_{min}$ to 200 $^{rev}/_{min}$ in 4.00 s. (a) Find the angular acceleration in $^{rev}/_{s^2}$ and the number of revolutions made by the motor in the 4.00-s interval. (b) How many more seconds are required for the fan to come to rest if the angular acceleration remains constant at the value calculated in part (a)?

5. A turntable rotates with a constant 2.25 $^{rad}/_{s^2}$ angular acceleration. After 4.00 s it has rotated through an angle of 60.0 rad. What was the angular velocity of the wheel at the beginning of the 4.00-s interval?

6. In a charming 19th century hotel, an old-style elevator is connected to a counterweight by a cable that passes over a rotating disk 2.50 m in diameter shown in fig below. The elevator is raised and lowered by turning the disk, and the cable does not slip on the rim of the disk but turns with it. (a) At how many rpm must the disk turn to raise the elevator at 25 $^{cm}/_{s}$?. (b) To start the elevator moving, it must be accelerated at $\frac{1}{8}g$. What must be the angular acceleration of the disk, in $^{rad}/_{s^2}$? (c) Through what angle (in radians and degrees) has the disk turned when it has raised the elevator 3.25 m between floors?



7. A wheel of diameter 40.0 cm starts from rest and rotates with a constant angular acceleration of 3.00 $^{rad}/_{s^2}$. At the instant the wheel has computed its second revolution, compute the radial acceleration of a point on the rim in two ways: (a) using the relationship $a_{rad} = \omega^2 r$ and (b) from the relationship $a_{rad} = {v^2}/{r}$.

8. You are to design a rotating cylindrical axle to lift 800-N buckets of cement from the ground to a rooftop 78.0 m above the ground. The buckets will be attached to a hook on the free end of a cable that wraps around the rim of the axle; as the axle turns, the buckets will rise. (a) What should the diameter of the axle be in order to raise the buckets at a steady 2.00 $^{cm}/_{s}$ when it is turning at 7.5 rpm? (b) If instead the axle must give the buckets an upward acceleration of 0.400 $^{m}/_{s^2}$, what should the angular acceleration of the axle be?

9. Small blocks, each with mass m, are clamped at the ends and at the center of a rod of length L and negligible mass. Compute the moment of inertia of the system about an axis perpendicular to the rod and passing through (a) the center of the rod and (b) a point one-fourth of the length from one end.

10. A uniform bar has two small balls glued to its ends. The bar is 2.00 m long and has mass 4.00 kg, while the balls each have mass 0.500 kg and can be treated as point masses. Find the moment of inertia of this combination about each of the following axes: (a) an axis perpendicular to the bar through its center; (b) an axis perpendicular to the bar through one of the balls; (c) an axis parallel to the bar through both balls; (d) an axis parallel to the bar and 0.500 m from it